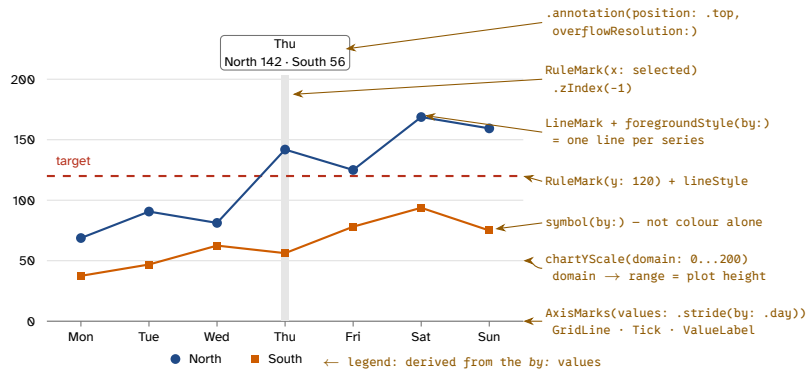
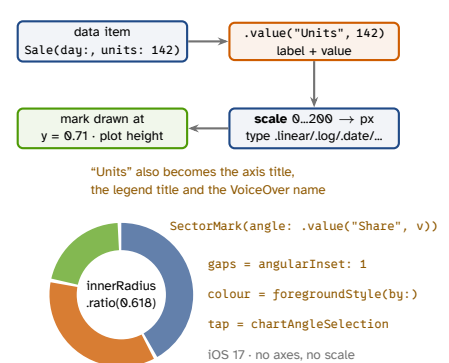


In one line: Swift Charts (iOS 16) is a **declarative grammar of graphics**: a Chart is a list of **marks** (one visual element per data item); each mark maps data to visual **properties** through `.value("Label", v)` **plottable values**; **scales** turn the data domain into pixels and colours, and **axes**, **legend**, VoiceOver and Audio Graphs are derived from the same labels — for free.

Anatomy — what each API draws



One property, end to end



The grammar

- `Chart(data) { BarMark(...) }` or `Chart { ForEach ... }`; a `@ChartContentBuilder`, so mark kinds mix (bars + a `RuleMark` average).
- Marks:** `Bar · Line · Point · Area · Rule · Rectangle` (heat map) · `Sector` (iOS 17). `yStart:yEnd: / xStart:xEnd:` draw ranges (Gantt, candles).
- Data kind picks the scale:** numbers = quantitative, `String/enum` = nominal (bands), `Date` = temporal; own types conform to `Plottable`. `.value("Day", d, unit: .day)` bins: the bar is one day wide.
- Encodings** `foregroundColorStyle / symbol / symbolSize / lineStyle(by:)`: each makes a scale *and* a legend entry. Bars on one x **stack** (stacking: `.normalized .center .unstacked`); `position(by:)` = side by side.
- Scales** `chartXScale(domain:range:type:): .linear .log .squareRoot .symmetricLog .power .date .category`. Bars include `0: .automatic(includesZero: false)` for a tight y. Fixed colours: `chartForegroundColorScale(["North": .blue])`.
- `chartYAxis(.hidden)`, `AxisMarks(position: .leading)`, `chartLegend(position:)`; `lines: interpolationMethod(.monotone .catmullRom .stepStart ...)`.
- `.annotation(position:alignment:spacing:)` pins any view to a mark; iOS 17 `overflowResolution: .init(x: .fit(to: .chart))` keeps it inside the chart.

Example — two series, target, tap-to-inspect, scroll

```
@State private var raw: Date? // raw x value, NOT a Sale
Chart {
  ForEach(sales) { s in
    LineMark(x: .value("Day", s.day, unit: .day),
             y: .value("Units", s.units))
    .foregroundColorStyle(by: .value("Store", s.store)) // = series
    .symbol(by: .value("Store", s.store))
    .interpolationMethod(.monotone) // no overshoot
  }
  RuleMark(y: .value("Target", 120))
  .lineStyle(StrokeStyle(lineWidth: 1, dash: [4, 3]))
  if let raw { RuleMark(x: .value("Day", raw, unit: .day)).zIndex(-1) }
}
.chartXSelection(value: $raw) // iOS 17
.chartScrollableAxes(.horizontal) // iOS 17
.chartXVisibleDomain(length: 3600 * 24 * 30) // SECONDS
```

Accessibility

VoiceOver per mark + Audio Graphs ("Describe chart", "Play Audio Graph") come free, named from the `.value` labels; refine with `.accessibilityLabel/Value` on the mark. A *hand-drawn* chart (Canvas, image) gets nothing: `.accessibilityChartDescriptor(r)` (iOS 15), `r: AXChartDescriptorRepresentable → AXChartDescriptor` (axes + series).

Interactivity and newer plots — by iOS version

```
16 chartOverlay { p in GeometryReader { g in ... } } + a
   DragGesture; p.value(atX: loc.x -
   g[p.plotAreaFrame].origin.x, as: Date.self).
17 chartXSelection(value:/range:), chartAngleSelection — gesture
   built in; custom: chartGesture { p in ...p.selectXValue(at:) };
   plotFrame replaces plotAreaFrame. Scroll: chartScrollableAxes,
   chartXVisibleDomain(length:), chartScrollPosition(x:),
   chartScrollTargetBehavior.
18 Vectorized: LinePlot AreaPlot BarPlot PointPlot
   RectanglePlot RulePlot SectorPlot — one plot per collection, key
   paths: LinePlot(sales, x: .value("Day", \.day), y:
   .value("Units", \.units)). Functions: LinePlot(x: "x", y:
   "y") { x in x * x }; return .nan = undefined.
26 Chart3D: PointMark(x:y:z:), SurfacePlot(x:y:z:){ x, z in
   ... } (y = f(x, z), y is up). chart3DPose($pose).
```

Performance

Each mark is laid out, hit-tested and exposed to VoiceOver — thousands in a `ForEach` crawl. **Aggregate first** (bin, min/max per pixel), off the main actor, never in body. WWDC24: **vectorized plots** for big uniformly-styled data, **marks** for per-item modifiers or `zIndex`.

Traps

- `chartXSelection` yields the **raw axis value** (a `Date` at 14:37), not an item — look up the item whose day contains it.
- `chartAngleSelection`: a point on the cumulative total — walk a running sum (*unverified*).
- Two series, no series:/by: → **one** zig-zag line joining both.
- `chartXVisibleDomain(length:)` on dates is in **seconds**.
- No unit: → hairline bars; `.catmullRom` overshoots below 0, `.monotone` not.
- Colour-only series fail colour-blind users — add `symbol(by:)`.
- Legend colours follow data order — pin them with `chartForegroundColorScale`.

Remember

"Mark × .value → scale → axis + legend + VoiceOver." 16 grammar · 17 select, scroll, pie · 18 vectorized + functions · 26 3D.